

Auction with Secret Reserve Price

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1 Model Outline

- One seller, N bidders.
- Seller's cost is either c_L or c_H with $\Pr(c_L) = \psi$.
- Buyer's value is i.i.d. distributed according to some F supported on $[c_L, 1]$, with density f .
- **First stage:** an English auction with initially hidden reserve price, which is revealed conditional on being met.
- **Second stage:** if the English auction fails, one bidder is randomly drawn to make a TIOLI (take-it-or-leave-it) offer to the seller. An individual bidder is drawn with probability $\delta \leq 1/N$.

Notation. $\underline{r}$ and $\bar{r}$: endogenous lower and upper support of the low-cost seller's mixed strategy (chosen in equilibrium, not drawn); r_H : high-cost seller's reserve; $\rho \in \Delta[\underline{r}, \bar{r}]$: cdf of the reserve r actually played on the path; β : symmetric stage-1 bidding strategy; $\tilde{\psi}$: belief that the seller is low-cost.

2 Preliminary Analysis

2.1 Second-stage belief and offer

If the chosen buyer has a belief $\tilde{\psi}$, he offers price c_H if and only if

$$v - c_H \geq \tilde{\psi}(v - c_L) \iff v \geq \frac{c_H - \tilde{\psi}c_L}{1 - \tilde{\psi}}. \quad (1)$$

2.2 Equilibrium structure

Consider an equilibrium where the high-cost seller plays a pure strategy r_H and the low-cost seller plays a mixed strategy denoted by $\rho \in \Delta[\underline{r}, \bar{r}]$.

Claim 1. $r_H > \bar{r}$.

Proof in Section 6. For any increasing bidding function, the optimal reserve price for a high-cost seller is strictly above that of a low-cost seller.

2.3 High-cost seller

When the bid exceeds $\bar{r}$ but the reserve is not met, bidders believe the seller is high-cost and bid according to $\beta(v) = v - \delta(v - c_H)$. Once a bidder has $\beta(v) \geq r_H$, the reserve is announced as met and people bid their own values. The high-cost seller's profit $\Pi_H(r_H)$ is

$$\Pi_H(r_H) = \int_{\beta^{-1}(r_H)}^1 \int_{c_L}^v (\max\{r_H, s\} - c_H) d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N. \quad (2)$$

Simplifying,

$$\begin{aligned}
\Pi_H(r_H) &= N \int_{\beta^{-1}(r_H)}^1 \left[(r_H - c_H) F(r_H)^{N-1} + \int_{r_H}^v (s - c_H) dF(s)^{N-1} \right] dF(v) \\
&= N \left[(r_H - c_H) F(r_H)^{N-1} (1 - F(\beta^{-1}(r_H))) + \int_{r_H}^1 (s - c_H) dF(s)^{N-1} \right. \\
&\quad \left. - F(\beta^{-1}(r_H)) \int_{r_H}^{\beta^{-1}(r_H)} (s - c_H) dF(s)^{N-1} - \int_{\beta^{-1}(r_H)}^1 F(v)(v - c_H) dF(v)^{N-1} \right].
\end{aligned} \tag{3}$$

The first-order condition is

$$r_H - c_H = (1 - \delta) \frac{1 - F\left(\frac{r_H - \delta c_H}{1 - \delta}\right)}{f\left(\frac{r_H - \delta c_H}{1 - \delta}\right)} - \frac{1}{F(r_H)^{N-1}} \int_{r_H}^{(r_H - \delta c_H)/(1 - \delta)} (v - c_H) dF(v)^{N-1}. \tag{4}$$

2.4 Belief after a failed English auction

If the English auction fails with final bid b , the buyer's belief is

$$\tilde{\psi} = \frac{\psi(1 - \rho(b))}{1 - \psi\rho(b)}. \tag{5}$$

3 Bidding in the First-Stage English Auction

Suppose a bidder with value v decides his bid by comparing $v - b$ against

$$\delta \max \left\{ v - c_H, \frac{\psi(1 - \rho(b))}{1 - \psi\rho(b)} (v - c_L), 0 \right\}. \tag{6}$$

His bid thus satisfies

$$b = v - \delta \max \left\{ v - c_H, \frac{\psi(1 - \rho(b))}{1 - \psi\rho(b)} (v - c_L), 0 \right\}. \tag{7}$$

The right-hand side is weakly decreasing in b and bounded between $[v - \delta \max\{v - c_H, \psi(v - c_L)\}, v - \delta \max\{v - c_H, 0\}]$. The bidding strategy β given ρ is thus

$$\beta(v) = \begin{cases} v - \delta \psi(v - c_L) & \text{if } v \leq \frac{r - \delta \psi c_L}{1 - \delta \psi}, \\ h^{-1}(v) & \text{if } v \in \left(\frac{r - \delta \psi c_L}{1 - \delta \psi}, v^* \right), \\ v - \delta(v - c_H) & \text{if } v \geq v^*, \end{cases} \tag{8}$$

where

$$h(b) \equiv \frac{(1 - \psi\rho(b))b - \delta\psi(1 - \rho(b))c_L}{1 - \delta\psi - (1 - \delta)\psi\rho(b)}, \tag{9}$$

i.e. $h(b)$ solves $v - b = \delta\tilde{\psi}(b)(v - c_L)$ with $\tilde{\psi}(b) = \psi(1 - \rho(b))/(1 - \psi\rho(b))$ from (5), and $v^* \in [(\underline{r} - \delta\psi c_L)/(1 - \delta\psi), (c_H - \psi c_L)/(1 - \psi)]$ is the solution to

$$v - c_H = \frac{\psi(1 - \rho((1 - \delta)v + \delta c_H))}{1 - \psi\rho((1 - \delta)v + \delta c_H)}(v - c_L) \iff \rho((1 - \delta)v + \delta c_H) = \frac{\psi(v - c_L) - (v - c_H)}{\psi(c_H - c_L)}. \quad (10)$$

4 Low-Cost Seller Profit

It suffices to focus on the low-type seller's profit. Suppose it sets reserve price $\underline{r}$.

4.1 When the auction fails

The English auction *fails* when every bidder stops before the reserve $\underline{r}$ is met, so the sale moves to stage 2. On the low bidding branch, failure is equivalent to $v_{\max} \leq z$ with

$$z \equiv \beta^{-1}(\underline{r}) = \frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}. \quad (11)$$

This event has probability $\Pr(\text{fail}) = F(z)^N$.

The seller may still earn positive profit in stage 2 if the selected buyer makes a c_H offer; each accepted c_H offer yields profit $c_H - c_L$. When the auction fails before $\underline{r}$, this event reveals no information about the seller's type, so the posterior belief that the seller is low-cost remains ψ . By (1) with this posterior, a buyer with value v offers c_H in stage 2 if and only if $v \geq v_\psi$, where

$$v_\psi \equiv \frac{c_H - \psi c_L}{1 - \psi}. \quad (12)$$

The subscript records that here $\tilde{\psi} = \psi$. Since failure implies $v_{\max} \leq z$, the seller earns this margin only when

$$v_\psi < v_{\max} < z \quad (\text{equivalently } z > v_{\max} > v_\psi).$$

Let $\underline{r}_0$ denote the reserve at which $z = v_\psi$,

$$\underline{r}_0 \equiv \delta\psi c_L + (1 - \delta\psi) \frac{c_H - \psi c_L}{1 - \psi} = \frac{(1 - \delta\psi)c_H - \psi(1 - \delta)c_L}{1 - \psi}. \quad (13)$$

Since z is increasing in $\underline{r}$, $v_\psi < v_{\max} < z$ is possible if and only if $\underline{r} > \underline{r}_0$; otherwise $\Pi_{\text{fail}} = 0$.¹

For $\underline{r} > \underline{r}_0$, one bidder is drawn to make a take-it-or-leave-it offer with probability δ . Here we distinguish between the highest bidder and the others, because stage 1 bidding reveals $v_{\max}$, which the seller observes, but not the other bidders' values.

If the selected bidder is the highest, his value equals $v_{\max}$, which the seller already knows, and if $v_{\max} \geq v_\psi$ he offers c_H . This contributes

$$\Pi_{\text{fail}}^{(1)} = \delta(c_H - c_L) \int_{v_\psi}^z dF(v)^N. \quad (14)$$

¹The cutoff $\underline{r}_0$ need not lie below the top of the value support. If buyers place sufficiently high prior probability on the low-cost type, v_ψ is so large that even the largest feasible $v_{\max}$ on a failure path need not warrant a c_H offer. Indeed, $\underline{r}_0 < 1$ if and only if $\psi < (1 - c_H)/(1 - c_L - \delta(c_H - c_L))$; if not, then $\underline{r}_0 \geq 1$ and $\Pi_{\text{fail}} = 0$ for every feasible reserve $\underline{r} \leq 1$.

If the selected bidder is not the highest, there are $N-1$ symmetric possibilities. The seller still knows $v_{\max}$ but not the selected value v , and a non-max bidder with value v offers c_H with probability $1 - F(v_\psi)/F(v)$. This contributes

$$\Pi_{\text{fail}}^{(2)} = \delta(N-1)(c_H - c_L) \int_{v_\psi}^z \left(1 - \frac{F(v_\psi)}{F(v)}\right) dF(v)^N. \quad (15)$$

Adding (14) and (15), and factoring out $\delta(c_H - c_L)$,

$$\Pi_{\text{fail}} = \Pi_{\text{fail}}^{(1)} + \Pi_{\text{fail}}^{(2)} = \delta(c_H - c_L) \left[\int_{v_\psi}^z dF(v)^N + (N-1) \int_{v_\psi}^z \left(1 - \frac{F(v_\psi)}{F(v)}\right) dF(v)^N \right]. \quad (16)$$

The first integral in (16) equals $F(z)^N - F(v_\psi)^N$. For $N \geq 2$, integration by parts on the second integral gives

$$\int_{v_\psi}^z \left(1 - \frac{F(v_\psi)}{F(v)}\right) dF(v)^N = \left(1 - \frac{F(v_\psi)}{F(z)}\right) F(z)^N - \frac{F(v_\psi)}{N-1} (F(z)^{N-1} - F(v_\psi)^{N-1}), \quad (17)$$

since $\int_{v_\psi}^z F(v)^{N-2} f(v) dv = (F(z)^{N-1} - F(v_\psi)^{N-1})/(N-1)$. Adding the two integrals in (16),

$$\begin{aligned} & \int_{v_\psi}^z dF(v)^N + (N-1) \int_{v_\psi}^z \left(1 - \frac{F(v_\psi)}{F(v)}\right) dF(v)^N \\ &= F(z)^N - F(v_\psi)^N + (N-1) \left(1 - \frac{F(v_\psi)}{F(z)}\right) F(z)^N - F(v_\psi) (F(z)^{N-1} - F(v_\psi)^{N-1}) \\ &= N(F(z) - F(v_\psi)) F(z)^{N-1}, \end{aligned} \quad (18)$$

where the second line uses $F(v_\psi)^N = F(v_\psi) F(v_\psi)^{N-1}$. When $N = 1$, only the first integral enters and (18) still holds. Substituting into (16),

$$\Pi_{\text{fail}} = \delta N (c_H - c_L) (F(z) - F(v_\psi)) F(z)^{N-1}. \quad (19)$$

Raising $\underline{r}$ makes stage 1 failure more likely because $F(z)^N$ increases, and for $\underline{r} > \underline{r}_0$ it also enlarges $[v_\psi, z]$ and raises the non-max offer probability $1 - F(v_\psi)/F(v)$.

4.2 When the auction succeeds

The auction *succeeds* when the reserve is met in stage 1, so the sale does not move to stage 2. This is equivalent to $v_{\max} > z$, with probability $1 - F(z)^N$. Once the reserve is met, bidders bid truthfully in the English auction. The highest-value bidder v wins and pays $\max\{\underline{r}, s\}$, so the seller's margin is $\max\{\underline{r}, s\} - c_L$. Averaging over the winner's value $v \geq z$ and the runner-up value $s \leq v$ gives

$$\Pi_{\text{succ}} = \int_z^1 \int_{c_L}^v (\max\{\underline{r}, s\} - c_L) d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N, \quad (20)$$

where $d(F(s)/F(v))^{N-1}$ is the distribution of s conditional on $v_{\max} = v$. Integrating by parts yields the equivalent form

$$\Pi_{\text{succ}} = N \left\{ (\underline{r} - c_L) F(\underline{r})^{N-1} (1 - F(z)) + \int_z^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \right\}. \quad (21)$$

The first term collects states where the runner-up's value is below $\underline{r}$, so the sale price equals the reserve. The double integral collects states where $s > \underline{r}$, so the winner pays the runner-up's value s (the usual second-price payment in an English auction).

4.3 Total profit and $\underline{r}^*$

Total profit is

$$\Pi(\underline{r}) = \mathbf{1}\{\underline{r} > \underline{r}_0\} \delta N(c_H - c_L) (F(z) - F(v_\psi)) F(z)^{N-1} + \Pi_{\text{succ}}, \quad (22)$$

The indicator $\mathbf{1}\{\underline{r} > \underline{r}_0\}$ makes explicit the reserve cutoff below which stage-2 profit is zero. Stage 2 offers always use the buyers' posterior belief $\tilde{\psi}$. In (22), failure before $\underline{r}$ leaves $\tilde{\psi} = \psi$, so the cutoff is v_ψ . If the seller deviates upward, stage-2 payoffs use the updated posterior $\tilde{\psi}$ from (5), which is strictly below ψ .

Claim 2. *The equilibrium lower support $\underline{r}^*$ is the argmax of $\Pi(\cdot)$ in (22).*

4.4 Proof of Claim 2

Proof. Let $\underline{r}^* \in \arg \max \Pi(\cdot)$. We show that any other lower support $\underline{r} \neq \underline{r}^*$ cannot be optimal.

Downward deviation. If $\underline{r} > \underline{r}^*$, failure still occurs before $\underline{r}$, so the posterior remains $\tilde{\psi} = \psi$ and (22) is exact, so $\Pi(\underline{r}^*) > \Pi(\underline{r})$.

Upward deviation. If $\underline{r} < \underline{r}^*$, announce $\underline{r}^*$ and write $z \equiv \beta^{-1}(\underline{r})$ and $z^* \equiv \beta^{-1}(\underline{r}^*)$.

Case 1: $\underline{r}^* > \underline{r} > \underline{r}_0$, so $z^* > z > v_\psi$. True failure profit is

$$\begin{aligned} \Pi_{\text{fail}}^{\text{true}} &= \delta N(c_H - c_L) (F(z) - F(v_\psi)) F(z)^{N-1} \\ &\quad + \delta(c_H - c_L) \int_z^{z^*} dF(v)^N + \delta(N-1)(c_H - c_L) \int_z^{z^*} \left(1 - \frac{F(v_\rho(v))}{F(v)}\right) dF(v)^N, \end{aligned} \quad (23)$$

where $v_\rho(v)$ is the c_H -offer cutoff with posterior belief $\tilde{\psi}(\beta(v))$ on $(z, z^*]$:

$$v_\rho(v) \equiv \frac{c_H - \tilde{\psi}(\beta(v)) c_L}{1 - \tilde{\psi}(\beta(v))} = \frac{(1 - \psi \rho(\beta(v))) c_H - \psi(1 - \rho(\beta(v))) c_L}{1 - \psi}.$$

On $(z, z^*]$, $\tilde{\psi}(\beta(v)) < \psi$ and $v_\rho(v) < v_\psi$. The first line of (23) is $\Pi_{\text{fail}}(\underline{r})$ from (19). The failure term in $\Pi(\underline{r}^*)$ is (16) with z^* in place of z ; splitting (16) at z into $[v_\psi, z]$ and $(z, z^*]$ and subtracting from (23) cancels the $\int_z^{z^*} dF(v)^N$ terms and gives

$$\Pi_{\text{fail}}^{\text{true}} - \Pi_{\text{fail}}(\underline{r}^*) = \delta(N-1)(c_H - c_L) \int_z^{z^*} \frac{F(v_\psi) - F(v_\rho(v))}{F(v)} dF(v)^N > 0$$

for $N \geq 2$, since $v_\rho(v) < v_\psi$ on $(z, z^*]$. Hence $\Pi_{\text{fail}}^{\text{true}} > \Pi_{\text{fail}}(\underline{r}^*)$. Success profit at $\underline{r}^*$ is unchanged, hence $\Pi_{\text{fail}}^{\text{true}} > \Pi(\underline{r}^*) > \Pi(\underline{r})$.

Case 2: $\underline{r}^* > \underline{r}$ but $\underline{r} \leq \underline{r}_0$. Then $\mathbf{1}\{\underline{r} > \underline{r}_0\} = 0$ and $\Pi_{\text{fail}} = 0$ at $\underline{r}$. If $\underline{r}^* > \underline{r}_0$, the integrals over $(z, z^*]$ in (23) are positive and $\Pi_{\text{fail}}^{\text{true}} > \Pi_{\text{fail}}(\underline{r}^*) = 0$; otherwise compare success profit only.

Conclusion. Either deviation is profitable, so $\underline{r}^*$ is the optimal lower support. $\square$

The low-cost seller's equilibrium payoff is $\Pi(\underline{r}^*)$. On the low branch, $\beta^{-1}(\underline{r}) = (\underline{r} - \delta\psi c_L)/(1 - \delta\psi)$.

5 Low-Cost Seller: Optimal Support and Mixed Strategy

This section completes the low-cost seller's problem. Claim 2 pins the equilibrium lower support to the reserve that maximizes $\Pi(\cdot)$ in (22). We characterize that maximizer, extend payoffs to reserves above the lower support on the mixing path, and construct the reserve distribution from the resulting indifference condition.

5.1 Optimal lower support

Subsection 4.3 and Claim 2 identify $\underline{r}^*$ as an argmax of $\Pi(\cdot)$ in (22). Write $z^* \equiv \beta^{-1}(\underline{r}^*)$ as in (11). Imposing $d\Pi(\underline{r})/d\underline{r} = 0$ at $\underline{r} = \underline{r}^*$ and rearranging yields

$$\begin{aligned} \underline{r}^* - c_L &= (1 - \delta\psi) \frac{1 - F(z^*)}{f(z^*)} \\ &+ \mathbf{1}\{\underline{r}^* > \underline{r}_0\} \frac{\delta(c_H - c_L)}{F(\underline{r}^*)^{N-1}} F(z^*)^{N-2} \left(NF(z^*) - (N-1)F(v_\psi) \right) \\ &- \frac{1}{F(\underline{r}^*)^{N-1}} \int_{\underline{r}^*}^{z^*} (s - c_L) dF(s)^{N-1}. \end{aligned} \quad (24)$$

Appendix A verifies this from (22).

5.2 Profit and indifference on the mixing path

In equilibrium the low-cost seller mixes a reserve $r \in [\underline{r}, \bar{r}]$ with cdf ρ , where $\underline{r}$ and $\bar{r}$ are the endogenous bounds of the support. If the realized reserve exceeds $\underline{r}$, write $z \equiv \beta^{-1}(\underline{r})$ and $z(r) \equiv \beta^{-1}(r)$ as in (11). Stage 1 failure can occur with $v_{\max}$ between z and $z(r)$; bids then reveal $\rho(\beta(v)) > 0$ and beliefs update via (5). On the failure path, the prior ψ applies below z and $\hat{\psi}(\beta(v))$ applies above it until $z(r)$. Payoff from announcing $r > \underline{r}$ is

$$\begin{aligned} \hat{\Pi}(r) &= \delta N(c_H - c_L) \left[(F(z) - F(v_\psi)) F(z)^{N-1} + F(z(r))^N - F(z)^N - N \int_z^{z(r)} F(v_\rho(v)) F(v)^{N-2} dF(v) \right] \\ &+ N \left\{ (r - c_L) F(r)^{N-1} (1 - F(\beta^{-1}(r))) \right. \\ &\quad \left. + \int_{\beta^{-1}(r)}^1 \int_r^v (s - c_L) dF(s)^{N-1} dF(v) \right\}. \end{aligned} \quad (25)$$

Appendix B derives the middle integral. When $r = \underline{r}$, the middle terms vanish and (25) coincides with (22).

On the equilibrium path $\underline{r} = \underline{r}^*$, so the seller is indifferent across the support:

$$\hat{\Pi}(r) = \Pi(\underline{r}^*) \quad \forall r \in [\underline{r}, \bar{r}]. \quad (26)$$

5.3 Construction of the mixing distribution

Differentiating (26) on the low branch yields an ordinary differential equation for $h(r) \equiv \beta^{-1}(r)$ and the mixing cdf $\rho(r)$; the algebra is in Appendix C. On the low branch, $h(r)$ coincides with (9) at $b = r$:

$$h(r) = \frac{(1 - \psi\rho(r))r - \delta\psi(1 - \rho(r))c_L}{1 - \delta\psi - (1 - \delta)\psi\rho(r)}, \quad (27)$$

with inverse

$$\rho(r) = \frac{-h(r) + r + \delta\psi(h(r) - c_L)}{\psi(-h(r) + r + \delta(h(r) - c_L))}. \quad (28)$$

Indifference on the low branch is equivalent to

$$h'(r) = \frac{F(r)^{N-1} (1 - F(h(r)))}{f(h(r))} \times \mathcal{B}(r, h(r))^{-1}, \quad (29)$$

where

$$\begin{aligned} \mathcal{B}(r, h) = & (h - c_L)F(h)^{N-1} - \int_r^h F(v)^{N-1} dv \\ & - \delta N(c_H - c_L) \left[F(h) - F\left(\frac{(1 - \psi\rho(r))c_H - \psi(1 - \rho(r))c_L}{1 - \psi}\right) \right] F(h)^{N-2}. \end{aligned} \quad (30)$$

Claim 3. Suppose F and f are continuous on $[c_L, 1]$ with $f > 0$. Let $\underline{r}^*$ satisfy (24), and set

$$h^* = \frac{\underline{r}^* - \delta\psi c_L}{1 - \delta\psi},$$

the inverse of the first bidding branch in (8). Then (29) admits a local solution $r \mapsto h(r)$ through $(\underline{r}^*, h^*)$.

Proof. At the lower end of the support, $\rho(\underline{r}^*) = 0$. Hence buyers use the prior belief ψ , bid according to $v - \delta\psi(v - c_L)$, and the inverse bid is the displayed h^* . Appendix C.3 shows that the denominator $\mathcal{B}(\underline{r}^*, h^*)$ is strictly positive. The right-hand side of (29) is therefore continuous around $(\underline{r}^*, h^*)$. Peano's theorem then gives a local solution. $\square$

A local solution is enough to define a mixed strategy on an interval $[\underline{r}^*, r_1]$: given h solving (29) on that interval, (28) pins down ρ , and (26) holds by construction on the low branch. For $r > \beta(v^*)$, $\beta^{-1}(r)$ is known in closed form and (54) in Appendix C extends ρ on the upper part of $[\underline{r}, \bar{r}]$. Together with $\bar{r} < r_H$ (Claim 1), this yields a mixed strategy equilibrium for the low-cost seller with payoff $\Pi(\underline{r}^*)$.

6 Upper Support and $\bar{r}$

At the top of the low-cost mixing support, $\rho(\bar{r}) = 1$. By (5), a stage 1 failure at bid $\bar{r}$ therefore leaves buyers with probability zero that the seller is low-cost: $\tilde{\psi}(\bar{r}) = 0$. By (1), a selected buyer offers c_H only if $v \geq c_H$.

6.1 Payoff at $r = \bar{r}$

Set $\underline{r} = \underline{r}^*$ on the equilibrium path and $r = \bar{r}$ in (25). Write $z^* \equiv \beta^{-1}(\underline{r}^*)$ and $z_{\bar{r}} \equiv \beta^{-1}(\bar{r})$. On the upper part of the support,

$$z_{\bar{r}} = \frac{\bar{r} - \delta c_H}{1 - \delta} \quad \text{when } \bar{r} > \beta(v^*), \quad (31)$$

as in (8) and Appendix C.4. Then

$$\begin{aligned} \hat{\Pi}(\bar{r}) = & \delta N(c_H - c_L) \left[\left(F(z^*) - F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right) F(z^*)^{N-1} \right. \\ & + F(z_{\bar{r}})^N - F(z^*)^N - N \int_{z^*}^{z_{\bar{r}}} F\left(\frac{(1 - \psi\rho(\beta(v)))c_H - \psi(1 - \rho(\beta(v)))c_L}{1 - \psi}\right) F(v)^{N-2} dF(v) \\ & + N \left\{ (\bar{r} - c_L) F(\bar{r})^{N-1} (1 - F(z_{\bar{r}})) \right. \\ & \left. \left. + \int_{z_{\bar{r}}}^1 \int_{\bar{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \right\} \right]. \end{aligned} \quad (32)$$

This is (25) evaluated at the upper support. The ψ block is the same as in (22) because it depends only on $\underline{r}^*$, not on $\bar{r}$. The remaining terms are the incremental failure and success payoffs from raising the reserve from $\underline{r}^*$ to $\bar{r}$.

At the upper endpoint of the failure integral, $\rho(\beta(z_{\bar{r}})) = \rho(\bar{r}) = 1$, so the integrand cutoff equals c_H :

$$\frac{(1 - \psi\rho(\bar{r}))c_H - \psi(1 - \rho(\bar{r}))c_L}{1 - \psi} = c_H.$$

On the equilibrium path, indifference gives

$$\hat{\Pi}(\bar{r}) = \Pi(\underline{r}^*) = \hat{\Pi}(\underline{r}^*), \quad (33)$$

where the second equality uses (25) at $r = \underline{r}^* = \underline{r}$.

6.2 Proof of Claim 1

Proof. The low-cost seller mixes on $[\underline{r}^*, \bar{r}]$ with $\rho(\bar{r}) = 1$. At bid $\bar{r}$, (5) gives $\tilde{\psi}(\bar{r}) = 0$, so buyers assign probability zero to the low type after failure at the top of the support. If $\bar{r} \geq r_H$, the high-cost seller's optimal pure reserve would lie inside (or below) the low-cost mixing support.

Consider the high-cost seller's problem in (3). Its optimal reserve r_H satisfies (4) and is strictly larger than the reserve that maximizes the low-cost *pure*-strategy profit $\Pi(\cdot)$ in (22): a high-cost seller faces a higher stage 2 cutoff and a steeper stage 1 bidding schedule $\beta(v) = v - \delta(v - c_H)$ than the low-cost branch $\beta(v) = v - \delta\psi(v - c_L)$ with $\psi < 1$. Hence $r_H > \arg \max_r \Pi(r) \geq \underline{r}^*$.

On the equilibrium path, the low-cost seller is indifferent across $[\underline{r}^*, \bar{r}]$, so $\hat{\Pi}(\bar{r}) = \Pi(\underline{r}^*)$ by (33). The payoff $\hat{\Pi}(\bar{r})$ in (32) is built from the low-cost bidding rule up to $z_{\bar{r}}$ in (31). If $\bar{r} \geq r_H$, the high-cost seller could meet reserve r_H while bidders above $\bar{r}$ already treat the seller as high-cost (Section 2), so the high-cost type's optimal separation reserve would not exceed the top of the low-cost support. That contradicts $r_H > \arg \max \Pi(\cdot)$ and the definition of $\bar{r}$ as the upper endpoint of the low-cost mixing support with $\rho(\bar{r}) = 1$.

Therefore $\bar{r} < r_H$. □

7 Case of Public Reserve Price

If the equilibrium is separating, the low-cost seller sets a reserve price given by

$$r_L - c_L = (1 - \delta) \frac{1 - F\left(\frac{r_L - \delta c_L}{1 - \delta}\right)}{f\left(\frac{r_L - \delta c_L}{1 - \delta}\right)} - \frac{1}{F(r_L)^{N-1}} \int_{r_L}^{(r_L - \delta c_L)/(1 - \delta)} (v - c_L) dF(v)^{N-1}. \quad (34)$$

whereas the high-cost seller sets reserve price weakly above r_H , solution to

$$r_H - c_H = (1 - \delta) \frac{1 - F\left(\frac{r_H - \delta c_H}{1 - \delta}\right)}{f\left(\frac{r_H - \delta c_H}{1 - \delta}\right)} - \frac{1}{F(r_H)^{N-1}} \int_{r_H}^{(r_H - \delta c_H)/(1 - \delta)} (v - c_H) dF(v)^{N-1}, \quad (35)$$

that is enough to deter low-type mimicking.

A FOC for $\underline{r}^*$

This appendix derives (24) from (22). Write $z \equiv \beta^{-1}(\underline{r})$ and v_ψ from (12). Total profit splits into the failure and success components (19) and (21):

$$\Pi(\underline{r}) = \mathbf{1}\{\underline{r} > \underline{r}_0\} \Pi_{\text{fail}}(\underline{r}) + \Pi_{\text{succ}}(\underline{r}), \quad (36)$$

where $\Pi_{\text{fail}} = 0$ for $\underline{r} \leq \underline{r}_0$. Differentiating (36) (holding fixed the region $\underline{r} > \underline{r}_0$ where the failure formula applies) gives

$$\frac{d\Pi(\underline{r})}{d\underline{r}} = \mathbf{1}\{\underline{r} > \underline{r}_0\} \frac{d\Pi_{\text{fail}}}{d\underline{r}} + \frac{d\Pi_{\text{succ}}}{d\underline{r}}. \quad (37)$$

A.1 Derivative of Π_{fail}

For $\underline{r} > \underline{r}_0$, Π_{fail} is (19). Since $z = (\underline{r} - \delta\psi c_L)/(1 - \delta\psi)$, $dz/d\underline{r} = 1/(1 - \delta\psi)$. Differentiating,

$$\begin{aligned} \frac{d\Pi_{\text{fail}}}{d\underline{r}} &= \frac{d}{d\underline{r}} [\delta N(c_H - c_L) (F(z) - F(v_\psi)) F(z)^{N-1}] \\ &= \delta N(c_H - c_L) \frac{1}{1 - \delta\psi} f(z) F(z)^{N-2} [NF(z) - (N-1)F(v_\psi)]. \end{aligned} \quad (38)$$

A.2 Derivative of Π_{succ}

Write Π_{succ} as in (21):

$$\Pi_{\text{succ}} = N \left\{ (\underline{r} - c_L) F(\underline{r})^{N-1} (1 - F(z)) + \int_z^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \right\}. \quad (39)$$

The outer bracket and the double integral are differentiated separately.

Double integral. Integration by parts gives

$$\begin{aligned} &\int_z^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \\ &= \int_{\underline{r}}^1 (s - c_L) dF(s)^{N-1} - F(z) \int_{\underline{r}}^z (s - c_L) dF(s)^{N-1} - \int_z^1 F(v) (v - c_L) dF(v)^{N-1}. \end{aligned} \quad (40)$$

Differentiating term by term yields (41)–(43) and the collapsed form

$$\frac{d}{d\underline{r}} \int_{\underline{r}}^1 (s - c_L) dF(s)^{N-1} = -(N-1)(\underline{r} - c_L) F(\underline{r})^{N-2} f(\underline{r}). \quad (41)$$

$$\begin{aligned} &\frac{d}{d\underline{r}} \left(F(z) \int_{\underline{r}}^z (s - c_L) dF(s)^{N-1} \right) \\ &= f(z) \frac{dz}{d\underline{r}} \int_{\underline{r}}^z (s - c_L) dF(s)^{N-1} \\ &\quad + (N-1)F(z) \left[(z - c_L) F(z)^{N-2} f(z) \frac{dz}{d\underline{r}} - (\underline{r} - c_L) F(\underline{r})^{N-2} f(\underline{r}) \right]. \end{aligned} \quad (42)$$

$$\frac{d}{d\underline{r}} \int_z^1 F(v)(v - c_L) dF(v)^{N-1} = -\frac{dz}{d\underline{r}}(N-1)(z - c_L)F(z)^{N-1}f(z). \quad (43)$$

Combining (41)–(43) gives

$$\begin{aligned} & \frac{d}{d\underline{r}} \int_z^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \\ &= -(N-1)(\underline{r} - c_L)F(\underline{r})^{N-2}f(\underline{r})(1 - F(z)) \\ & \quad - f(z) \frac{d\beta^{-1}(\underline{r})}{d\underline{r}} \int_{\underline{r}}^z (s - c_L) dF(s)^{N-1}. \end{aligned} \quad (44)$$

Bracket term.

$$\begin{aligned} & \frac{d}{d\underline{r}} [(r - c_L)F(\underline{r})^{N-1}(1 - F(z))] \\ &= \left[F(\underline{r})(1 - F(z)) \right. \\ & \quad \left. + (N-1)(\underline{r} - c_L)(1 - F(z))f(\underline{r}) \right] F(\underline{r})^{N-2} \\ & \quad - (\underline{r} - c_L)F(\underline{r})f(z) \frac{1}{1 - \delta\psi} \end{aligned} \quad (45)$$

Multiplying (45) and (44) by N gives $d\Pi_{\text{succ}}/d\underline{r}$. The $(N-1)(\underline{r} - c_L)$ terms cancel, leaving

$$\begin{aligned} \frac{d\Pi_{\text{succ}}}{d\underline{r}} &= NF(\underline{r})^{N-2} \left[F(\underline{r})(1 - F(z)) - (\underline{r} - c_L)F(\underline{r})f(z) \frac{dz}{d\underline{r}} \right] \\ & \quad - Nf(z) \frac{dz}{d\underline{r}} \int_{\underline{r}}^z (s - c_L) dF(s)^{N-1}. \end{aligned} \quad (46)$$

A.3 Full FOC and simplification

Substituting (38) and (46) into (37) and setting $d\Pi(\underline{r})/d\underline{r} = 0$ at $\underline{r} = \underline{r}^*$ gives

$$\begin{aligned} 0 &= \mathbf{1}\{\underline{r}^* > \underline{r}_0\} \delta N(c_H - c_L) \frac{1}{1 - \delta\psi} f(z^*) F(z^*)^{N-2} \left[NF(z^*) - (N-1)F(v_\psi) \right] \\ & \quad + NF(\underline{r}^*)^{N-2} \left[F(\underline{r}^*)(1 - F(z^*)) - (\underline{r}^* - c_L)F(\underline{r}^*)f(z^*) \frac{dz}{d\underline{r}} \right] \\ & \quad - Nf(z^*) \frac{dz}{d\underline{r}} \int_{\underline{r}^*}^{z^*} (s - c_L) dF(s)^{N-1}. \end{aligned} \quad (47)$$

Collecting the $(\underline{r}^* - c_L)$ terms on the left and dividing by $NF(\underline{r}^*)^{N-1}f(z^*)dz/d\underline{r}$ yields (24). Explicitly, (47) is equivalent to

$$NF(\underline{r}^*)^{N-1}f(z^*) \frac{dz}{d\underline{r}} (\underline{r}^* - c_L) = NF(\underline{r}^*)^{N-1}(1 - F(z^*)) - Nf(z^*) \frac{dz}{d\underline{r}} \int_{\underline{r}^*}^{z^*} (s - c_L) dF(s)^{N-1} + \mathbf{1}\{\underline{r}^* > \underline{r}_0\} \frac{d\Pi_{\text{fail}}}{d\underline{r}}, \quad (48)$$

so

$$\underline{r}^* - c_L = \frac{1 - F(z^*)}{f(z^*)dz/d\underline{r}} + \mathbf{1}\{\underline{r}^* > \underline{r}_0\} \frac{d\Pi_{\text{fail}}/d\underline{r}}{NF(\underline{r}^*)^{N-1}f(z^*)dz/d\underline{r}} - \frac{1}{F(\underline{r}^*)^{N-1}} \int_{\underline{r}^*}^{z^*} (s - c_L) dF(s)^{N-1}, \quad (49)$$

which is (24). Substituting (49) back into (47) shows $d\Pi(\underline{r}^*)/d\underline{r} = 0$ if and only if (24) holds:

$$\frac{d\Pi}{d\underline{r}} = -NF(\underline{r})^{N-1}f(z)\frac{dz}{d\underline{r}}\left[(\underline{r} - c_L) - \text{RHS of (24)}\right]. \quad (50)$$

The first line on the right-hand side of (24) comes from the success bracket in (46) and is present whether or not $\underline{r}^* > \underline{r}_0$. The failure adjustment (second line) comes from (38). The integral is the success double-integral term.

Algebraic check. When $\mathbf{1}\{\underline{r}^* > \underline{r}_0\} = 1$, all three lines contribute. When $\mathbf{1}\{\underline{r}^* > \underline{r}_0\} = 0$, (24) reduces to the first-order condition from $d\Pi_{\text{succ}}/d\underline{r} = 0$ alone.

B Profit for $r > \underline{r}$

Write $z \equiv \beta^{-1}(\underline{r})$ and $z(r) \equiv \beta^{-1}(r)$ as in (11), v_ψ from (12), and $v_\rho(v)$ as in (23). The stage-2 failure block in (25) is

$$\begin{aligned} & \delta N(c_H - c_L) (F(z) - F(v_\psi)) F(z)^{N-1} \\ & + \delta N(c_H - c_L) \int_z^{z(r)} \left(1 - \frac{F(v_\rho(v))}{F(v)}\right) dF(v)^N, \end{aligned} \quad (51)$$

plus the stage-1 success terms in (25). The first line is (19) at $\underline{r}$. The integral over $(z, z(r)]$ collects updated-belief payoffs as in the $(z, z^*]$ terms of (23) with $\underline{r}^*$ replaced by r . Since $\int (1 - F_\rho/F) dF^N = N \int (F - F_\rho) F^{N-2} dF$ with $F_\rho \equiv F(v_\rho(v))$,

$$\int_z^{z(r)} \left(1 - \frac{F_\rho}{F}\right) dF^N = F(z(r))^N - F(z)^N - N \int_z^{z(r)} F_\rho F^{N-2} dF,$$

which yields the middle terms in (25).

C Mixed-strategy indifference and the ODE

C.1 Incremental indifference

On $[\underline{r}, \bar{r}]$, equilibrium requires $\hat{\Pi}(r) = \Pi(\underline{r}^*)$. The ψ block in (25) does not depend on r ; with $\underline{r} = \underline{r}^*$ on path,

$$\begin{aligned} & \delta N(c_H - c_L) \left[F(\beta^{-1}(r))^N - F(\beta^{-1}(\underline{r}^*))^N - N \int_{\beta^{-1}(\underline{r}^*)}^{\beta^{-1}(r)} F \left(\frac{(1 - \psi \rho(\beta(v)))c_H - \psi(1 - \rho(\beta(v)))c_L}{1 - \psi} \right) F(v)^{N-2} dF(v) \right] \\ & + N \left\{ (r - c_L) F(r)^{N-1} (1 - F(\beta^{-1}(r))) + \int_{\beta^{-1}(r)}^1 \int_r^v (s - c_L) dF(s)^{N-1} dF(v) \right\} \\ & = N \left\{ (\underline{r}^* - c_L) F(\underline{r}^*)^{N-1} (1 - F(\beta^{-1}(\underline{r}^*))) + \int_{\beta^{-1}(\underline{r}^*)}^1 \int_{\underline{r}^*}^v (s - c_L) dF(s)^{N-1} dF(v) \right\}. \end{aligned} \quad (52)$$

C.2 Derivative and the ODE

On the low branch, $\beta^{-1}(r) = h(r)$. Differentiating (52) in r gives

$$\begin{aligned}
0 = \delta N(c_H - c_L) & \left[F(h(r))^{N-1} f(h(r)) h'(r) - F\left(\frac{(1 - \psi\rho(r))c_H - \psi(1 - \rho(r))c_L}{1 - \psi}\right) F(h(r))^{N-2} f(h(r)) h'(r) \right] \\
& + F(r)^{N-1} (1 - F(h(r))) - (r - c_L) F(r)^{N-1} f(h(r)) h'(r) \\
& - f(h(r)) h'(r) \int_r^{h(r)} (s - c_L) dF(s)^{N-1}.
\end{aligned} \tag{53}$$

Collecting terms with $h'(r)$ in (53) and solving gives (29)–(30).

C.3 Initial denominator

At the lower support, $\rho(\underline{r}^*) = 0$, so buyers keep the prior belief ψ after failure. The first bidding branch in (8) therefore gives

$$\beta^{-1}(\underline{r}^*) = \frac{\underline{r}^* - \delta\psi c_L}{1 - \delta\psi}.$$

The stage-2 cutoff in (30) also reduces to $(c_H - \psi c_L)/(1 - \psi)$. To evaluate the ODE denominator at the initial point, first use

$$\begin{aligned}
& \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)} (s - c_L) dF(s)^{N-1} \\
& = (\beta^{-1}(\underline{r}^*) - c_L) F(\beta^{-1}(\underline{r}^*))^{N-1} - (\underline{r}^* - c_L) F(\underline{r}^*)^{N-1} - \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)} F(v)^{N-1} dv.
\end{aligned}$$

Substituting this identity into (30) and using (24) gives

$$\begin{aligned}
& \mathcal{B}(\underline{r}^*, \beta^{-1}(\underline{r}^*)) \\
& = (1 - \delta\psi) \frac{1 - F(\beta^{-1}(\underline{r}^*))}{f(\beta^{-1}(\underline{r}^*))} F(\underline{r}^*)^{N-1} \\
& \quad + \delta(c_H - c_L) F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) F(\beta^{-1}(\underline{r}^*))^{N-2}.
\end{aligned}$$

This is the initial denominator used in Claim 3. It is strictly positive whenever $F(\beta^{-1}(\underline{r}^*)) < 1$.

C.4 High reserves

For $r > \beta(v^*)$, $\beta^{-1}(r) = (r - \delta c_H)/(1 - \delta)$ and (53) becomes

$$\begin{aligned}
0 = \delta N(c_H - c_L) & \left(F\left(\frac{r - \delta c_H}{1 - \delta}\right) - F\left(\frac{(1 - \psi\rho(r))c_H - \psi(1 - \rho(r))c_L}{1 - \psi}\right) \right) F\left(\frac{r - \delta c_H}{1 - \delta}\right)^{N-2} f\left(\frac{r - \delta c_H}{1 - \delta}\right) \frac{1}{1 - \delta} \\
& + F(r)^{N-1} \left(1 - F\left(\frac{r - \delta c_H}{1 - \delta}\right) \right) \\
& - \frac{1}{1 - \delta} f\left(\frac{r - \delta c_H}{1 - \delta}\right) \left[\left(\frac{r - \delta c_H}{1 - \delta} - c_L \right) F\left(\frac{r - \delta c_H}{1 - \delta}\right)^{N-1} - \int_r^{(r - \delta c_H)/(1 - \delta)} F(v)^{N-1} dv \right],
\end{aligned} \tag{54}$$

which pins down $\rho(r)$ on the upper part of the support.